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Development and validation of graphics-based digital twin framework for UAV-aided post-earthquake inspection of high-rise buildings

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Summary

Traditional post-earthquake inspection of civil infrastructure is conducted manually, taking a considerable amount of time and often putting inspectors in harm's way. This problem is exacerbated in modern cities, where millions of people can be left homeless until their residences are deemed safe to reinhabit. Image collection enabled by commercial unmanned aerial vehicles (UAVs) combined with computer vision-based techniques has provided an alternative with high potential for rapid post-earthquake inspection. However, the extracted images of the damage alone are inadequate to evaluate the system-level safety condition of a structure. The quality of the visual information also heavily relies on the effectiveness of the UAV inspection scheme which is susceptible to environmental uncertainties. To this end, a graphics-based digital twin (GBDT) framework is developed for UAV-aided post-earthquake inspection of high-rise buildings and validated using a high-rise building in Guangzhou, China. The GBDT is comprised of a finite element (FE) model and a photorealistic computer graphics (CG) model, with the latter being informed by the former, jointly providing as a comprehensive virtual representation of the structure so that every step of the post-earthquake inspection procedure can be planned and evaluated virtually. First, to avoid the cumbersome nature of constructing the graphical representation of the numerous components in high-rise buildings, the CG model in the GBDT is created by automatically importing structural components from the FE model and adding nonstructural components according to the dimensions of the as-built structure. This fast modeling process as well as the accuracy of the virtual presentation are validated by point cloud comparisons between the CG model and the as-built structure. Subsequently, the GBDT is used to showcase the evaluation of UAV flight schemes for post-earthquake inspection of high-rise buildings. To shorten flight time and place more emphasis on potential damage, FE analysis is conducted to determine the earthquake-induced damage locations. Consistent damage hotspots are then

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marked on the CG model, along with restrictions from the real environment such as obstacles, weak satellite signal, wind speed, and lighting conditions considered in the synthetic environment. Finally, applying the synthetic environment as the testbed, three UAV-aided inspection schemes are implemented virtually and the best UAV flight scheme is determined for the assumed field inspection. This example demonstrates the flexibility of the GBDT in representing the real-world structure and environmental conditions and its efficacy in assisting decision making for rapid and effective structural inspection in the aftermath of an earthquake.

KEYWORDS
3D synthetic environment, earthquake engineering, graphics-based digital twin (GBDT), high-rise buildings, post-earthquake inspection, UAV-aided inspection

1 | INTRODUCTION

Earthquakes are among the most catastrophic natural disasters faced by modern societies. Damage to civil infrastructure is inevitable due to the rapid urbanization in regions of high seismicity. Rapid inspection and evaluation of damaged structures therefore become crucial and urgent to ensure the safety of life and property, as well as to restore normalcy to society. Traditional post-earthquake inspections are usually conducted by structural engineering experts based on visual checks and empirical judgment. For example, during the 1978 Thessaloniki earthquake in Greece, it took almost 1000 engineers about 2 months to inspect more than 150,000 apartments in 30,000 buildings within Thessaloniki city and 30,000 apartments in 13,000 buildings in the surrounding rural area. This earthquake, along with the subsequent 1981 Aklyonides–Korinthos earthquake, led to the establishment of the Earthquake Planning and Protection Organization of Greece in 1983, which marked the beginning of systematic post-earthquake inspection guidelines.[1] Similarly, the U.S. Applied Technology Council published a field manual for post-earthquake safety evaluation of buildings in 1989.[2] In 1997, China passed the law on Earthquake Prevention and Disaster Reduction[3] which provided similar guidelines. Despite the standardization of post-earthquake inspection procedures by these manuals and laws, inspections are still conducted manually, and therefore are incredibly time-consuming and perilous for inspectors, especially when dealing with high-rise buildings. Because of the significant heights and dimensions of such buildings, damage to the upper floors is nearly impossible to identify from the ground and thus requires extensive inspection time inside the damaged building. Moreover, high-rise buildings play a major role in sheltering large populations in cities; until these buildings are inspected, the residents are essentially homeless. Thus, means for rapid post-earthquake inspection of the civil infrastructure is direly needed.

With the recent development of the drone industry,[4–7] image collection enabled by commercial unmanned aerial vehicles (UAVs) provides a significant opportunity for rapid post-earthquake inspections. Researchers have proposed that UAV-collected images can be utilized to build computer graphics (CG) models with point cloud reconstruction techniques,[8,9] which can then be used for visual inspection. The capacity and effectiveness of the inspection can be further facilitated by automated damage detection based on the UAV-collected images. Early research about damage detection techniques employed heuristic feature extraction methods, applying a threshold or a classifier to the output of a manually designed filter for a particular type of damage.[10–13] To address the challenges faced by such heuristic feature extraction techniques, such as complex design, narrow applicability, and low accuracy, researchers have turned toward machine learning-based methods.[14,15] Machine learning-based algorithms can handle several types of damage simultaneously. Damage can also be projected onto a mesh model to better guide engineers in assessing structural conditions. For example, Hoskere et al.[16] generated a 3D mesh model from UAV-collected images of a damaged building following the 2017 Central Mexico Earthquake. By projecting the inferred damage information onto the 3D mesh model, a condition-aware model was developed for damage documentation and higher-level inspection decisions. However, such drone surveys still need to be flown by human operators in the field.

The Resilient City initiative developed by the San Francisco Bay Area Planning and Urban Research Association (SPUR) is aimed at developing customized inspection programs and pre-assess structures to enable rapid post-earthquake response.[17] While the Resilient City initiative is designed to allow building owners to better respond to seismic disasters, human inspectors are still relied upon.

Inspired by the Resilient City Initiative, Wang et al.[18] proposed a graphics-based digital twin (GBDT) framework for automated computer vision-based post-earthquake inspection and evaluation of as-built structures. The concept of the digital twin was first introduced in the early 2000s[19] and has since expanded to the field of civil engineering, encompassing the monitoring and management of construction[20–22] and the operation and maintenance of civil infrastructure.[23–25] The digital twin, continuously updated by the sensor information and load history from the physical world, provides predictive information about the physical entity and demonstrates the potential for improved understanding of the physical world. However, much of the information about the current state of a structure is visual in nature. Thus, the GBDT builds on the work by Hoskere et al.[26–28]

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and Narazaki et al.[29–31] on physics-based graphics models (PBGM) and incorporates a photorealistic graphics model with the traditional digital twin. As shown in Figure 1,[18] the "digital twin" in the GBDT consists of two components, which are as follows: (i) a global FE model and (ii) an associated photorealistic CG model of the as-built structure. Note that the CG model can realistically represent the appearance of the as-built structure in both undamaged and damaged conditions. Following the steps outlined in Figure 1, the GBDT of a simple five-story reinforced concrete benchmark building was built and various UAV inspection trajectories for post-earthquake scenarios were designed and assessed in the synthetic environment.[18] However, the application of GBDT has yet to be applied in the field, where structures such as high-rise buildings in densely populated areas pose significant additional challenges. Moreover, the quality of UAV-collected images is highly dependent on various environmental factors, such as weather, lighting, satellite signals, and obstacles. Therefore, the environmental conditions at the site of the target structure, in addition to the structure itself, should also be replicated in the visual representation, which was not considered in previous studies.

To validate the GBDT framework using real-world structures and account for the environmental restrictions in the context of post-earthquake inspection, this paper concentrates on a 45-floor building located in Guangzhou, China. In contrast to the fictional low-rise building with a regular plan examined in the previous study,[18] the structure chosen for this investigation is a high-rise building characterized by a more intricate architectural form. Following the research overview in Figure 2, the FE model is first built according to the structural drawings, providing meshes of the structural components that can be automatically imported into the CG software. The nonstructural components are then added to the CG model based on the given dimensions from the architectural drawings. The exterior surfaces of the CG model are textured using the photos taken on-site (Section 2). Based on the undamaged CG model, appropriate parameters are determined for the UAV flight path with the aim of constructing a high-quality point cloud model of the as-built structure. This point cloud model is then used to validate the fast CG modeling

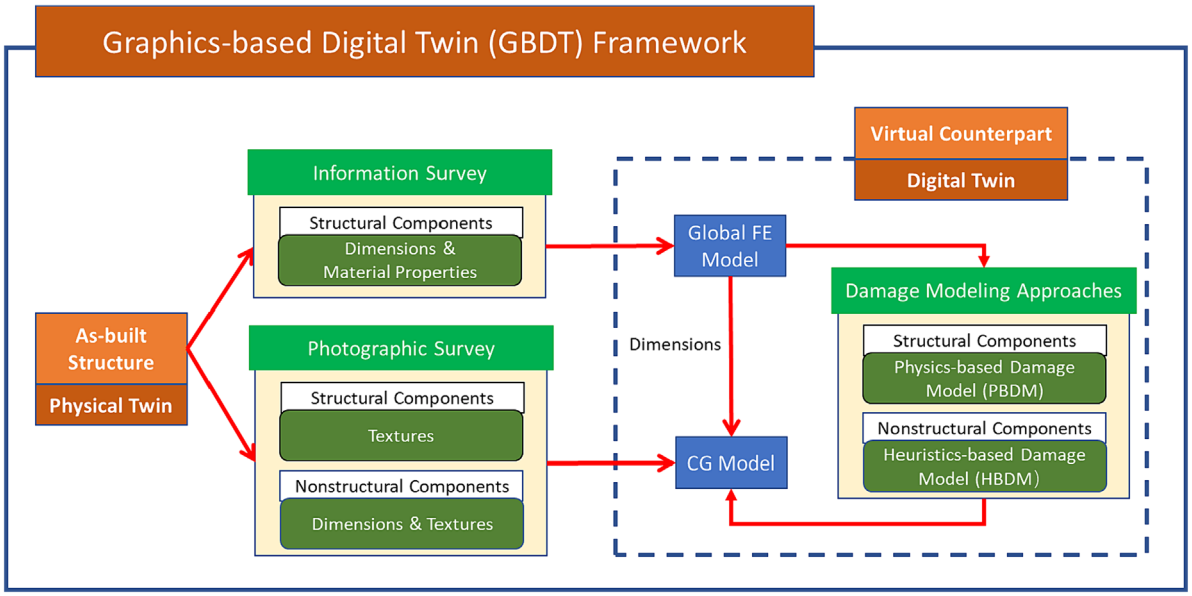


FIGURE 1 Graphics-based digital twin (GBDT) Framework.[18]

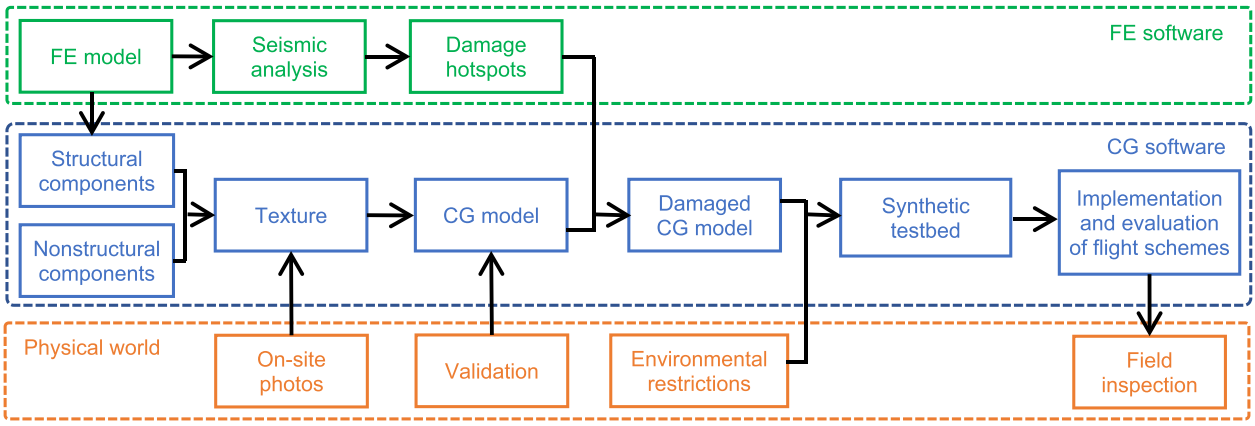


FIGURE 2 Research overview.

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process and accuracy of the virtual representation (Section 3), which is carried out on a real-world high-rise structure for the first time. For the damaged CG model, nonlinear analysis using the global FE model under assumed seismic excitation is conducted to determine the location and extent of damage. With this information, damage hotspots are marked in the CG model to highlight areas that require special attention during post-earthquake inspections (Section 4). Additionally, environmental restrictions including obstacles, weak satellite signal, wind speed, and lighting conditions are considered to create a synthetic testbed for UAV flight schemes. Subsequently, three UAV-aided inspection schemes are implemented virtually, and the best UAV flight scheme is determined for the planned field inspection (Section 5). This example demonstrates the flexibility of the GBDT in representing the real-world structure and environmental conditions, as well as the significant potential of the GBDT framework for UAV-aided post-earthquake inspection of high-rise buildings.

2 | DEVELOPMENT OF THE GBDT

This section provides the steps to create the GBDT of the high-rise building, including FE modeling, CG modeling, and the environment in which the building is located.

2.1 | Target structure

The target structure of this study is a residential building located in Guangzhou, China. This 45-floor high-rise building, which is one of two twin towers, is a reinforced concrete shear wall structure shown in Figure 3a. The dimension of the structural plan is 25.0 m × 27.2 m, and the total height is 138 m, including 6 m for the ground floor, and 3 m for each of the remaining floors. The concrete tensile strength of the shear walls on floors 1–8, 9–21, and 22–45 are 2.04, 1.89, and 1.71 N/mm^2, respectively, and the concrete tensile strength of beams and slabs is 1.43 N/mm^2. The layout of the structure is shown in Figure 3b.

2.2 | FE model of the structure

The FE model is a key component of the GBDT, as it not only provides information on structural components for the CG model but also is used to predict the damage through nonlinear time history analysis. In this study, the structural analysis software SAP2000 is adopted to build the nonlinear FE model of the target structure. Structural components including the beams, columns, slabs, and shear walls are constructed in the FE model. The beams and columns are modeled as frame elements, the slabs as thick shells, and the shear walls as nonlinear layered shells. Three types of nonlinearity are considered for the FE model in this study, including the P–Δ effect for the overall structure, the layered shell arrangement for the shear walls, and the plastic hinge setting for the beams. The plastic hinges are placed at one-tenth locations near the two ends of the beams. Modal analysis is conducted to examine the natural periods and mode shapes of the structure. The first three natural periods and

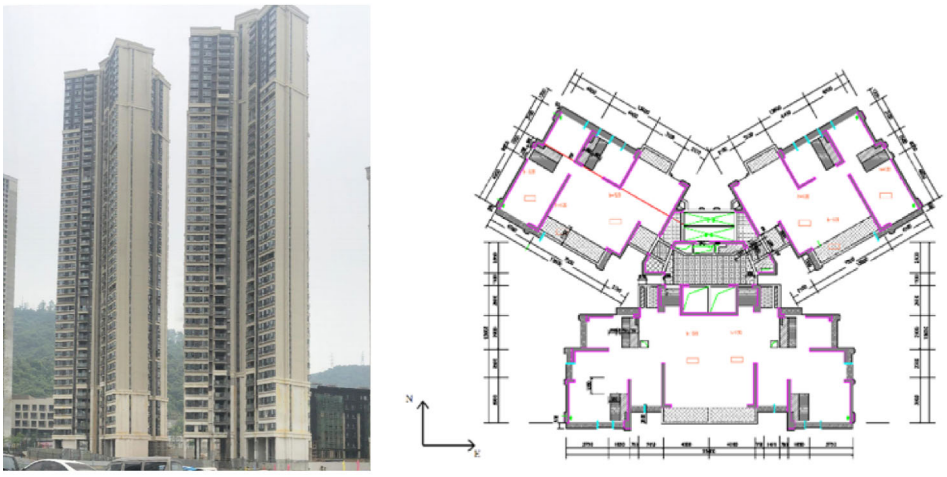


FIGURE 3 Target structure: (a) photo (the left building) and (b) plan.

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the corresponding first three mode shapes are translation in the north–south direction (3.46 s), translation in the east–west direction (3.08 s), and torsion (2.62 s), as shown in Figure 4. The modal results obtained are deemed reasonable for a typical high-rise building. In practical applications, the natural frequencies and other modal characteristics of the as-built structure can be identified through field tests, offering a viable means for the validation of the FE model.

2.3 | CG model of the structure

This section describes the construction of both the structural and nonstructural components of the CG model. The structural components are automatically built based on the FE model, whereas the nonstructural components are manually built in accordance with the architectural drawings. The CG model is then textured using the on-site photos of the real building.

2.3.1 | Structural components of the CG model

The open-source 3D software, Blender, is adopted to build the CG model of the target structure. In the CG model, the structural components are generated automatically based on the mesh information of the FE model in two steps. First, the mesh information of the FE model, including the node coordinates, shell element nodes, and frame nodes, is exported and saved using the SAP-Python Application Programming Interface (API); next, the saved mesh information is imported into the Blender environment using the Blender Python API. The structural components of the CG model generated by the above process are shown in Figure 5a.

2.3.2 | Nonstructural components of the CG model

The nonstructural components on the exterior surfaces of the high-rise structure include the balconies, glass windows, and shutters. Since these nonstructural components are configured in different ways, modeling with the Blender Python API would be time-consuming and labor-intensive. Therefore, this study uses the graphical user interface (GUI) to manually build the nonstructural components of the CG model. Cube units are created and reshaped in Blender according to the dimensions in the architecture drawings to obtain a refined 3D replica of the nonstructural component. Figure 6a shows the 3D model of the balcony part by the above method. The finished nonstructural components are then assembled into the structural components to obtain a complete CG model of the high-rise structure (see Figure 5b).

2.3.3 | Texture of the CG model

The textures of the CG model are generated based on photos of the exterior surface of the high-rise structure taken in the field. Four types of textures are prepared including the base color map, displacement map, normal map, and roughness map. Those textures are combined through

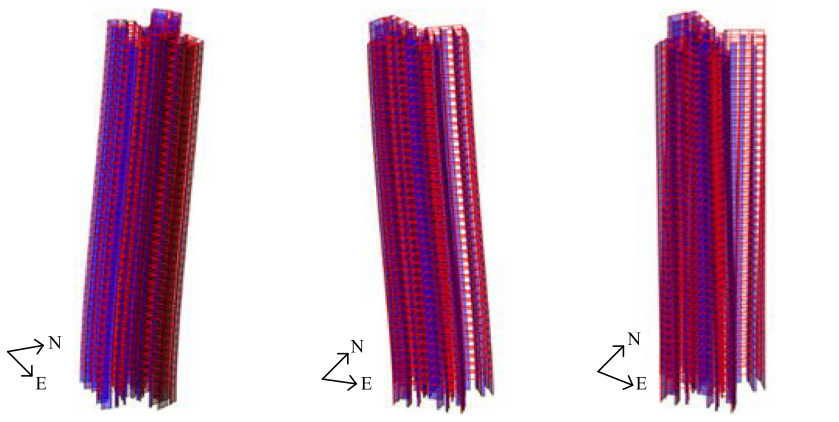


FIGURE 4 Mode shapes of the structure.

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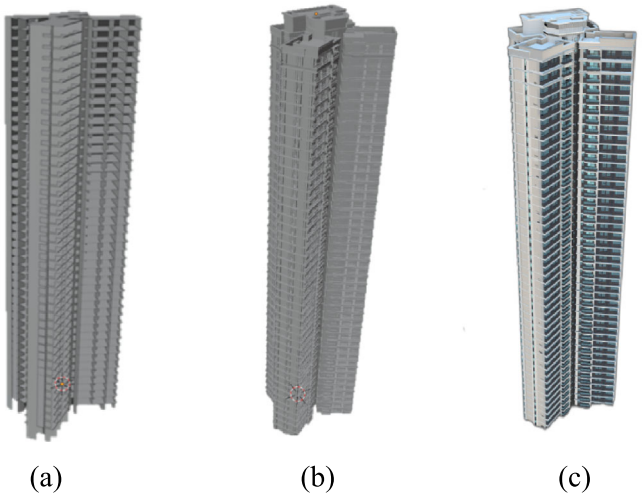


FIGURE 5 CG model of the structure: (a) structural components only, (b) structural and nonstructural components, and (c) textured.

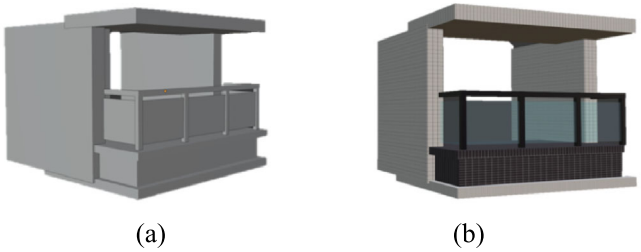


FIGURE 6 Balcony in the CG model: (a) untextured and (b) textured.



FIGURE 7 Visual comparison: (a) real structure and (b) CG model.

the principled bidirectional scattering distribution function (BSDF) shader in Blender. Nodes are added to the principled BSDF shader to further adjust the base color and the brightness intensity of the textures so that the CG model can resemble the real structure as much as possible. The structure and balcony part after texturing are shown in Figures 5c and 6b, respectively. Figure 7 compares the photo of the real high-rise structure and a screenshot of the final CG model from a similar pose. A high degree of similarity can be seen between these two images.

2.4 | Environmental adjustment for the CG model

To conduct practical post-earthquake UAV flight planning using the CG model, the surrounding environment of the target structure also should be considered. Roads, trees, and especially adjacent buildings that must be respected in determining the UAV path are constructed in Blender according to the physical environment of the target structure. The locations and dimensions of surrounding buildings are obtained by satellite images, while the dimensions and heights of the adjacent high-rise buildings are read from drawings. The CG models of the surrounding buildings are created by performing manual operations on cube units using the GUI. Because the UAV flight path only concerns with the positions and sizes

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of the obstacles, there is no need to texture the surrounding buildings. Roads are built by scaling the plane model and adjusting the terrain height and size. Trees are acquired from the open-source asset library and placed referring to the buildings and roads. Finally, the high-dynamic range image environment map is used to add lighting to the CG environment. The CG model of the high-rise structure with those environmental considerations is shown in Figure 8.

3 | VALIDATION OF THE CG MODEL

In this section, an issue that arises during the field UAV inspection of the as-built structure is first described. Subsequently, a solution is sought based on the analysis of the synthetic environment using the CG model of the high-rise building. An updated UAV flight path is then implemented on site, and a high-quality point cloud model of the as-built structure is constructed, which serves as a reference for validation of the CG model.

3.1 | Initial point cloud model of the structure

In the context of post-earthquake UAV-aided inspection of engineering structures, a point cloud model built using the UAV-collected images provides a means to evaluate the effectiveness of the flight path. Typically, a higher point cloud density indicates a higher quality model and thus a more effective UAV flight path.

In this example, the images of the as-built structure were captured by a commercial UAV DJI Phantom 4 Pro V2.0 with parallel trajectory loops. The detailed parameters of this UAV are shown in Table 1.[32] The images were initially taken at an interval of 5 m, both vertically and horizontally, with a downward camera angle of 30°, and at a distance of 5 m away from the building surfaces. The images were then used to construct the point cloud model of the as-built structure in the photogrammetry software RealityCapture. However, the resulting point cloud model shown in Figure 9a was of poor quality, as evident from large missing areas and a low point cloud density assessed by the software CloudCompare shown in Figure 9c. To show the details, only three stories are displayed.

To obtain deeper insight into the encountered issue, the UAV survey was replicated in the synthetic environment. The camera poses along the UAV trajectory relative to the building surfaces were obtained directly from the path planning software. Synthetic cameras were placed at those locations to take pictures with the same camera angle and image resolution as in the field. These synthetic images were then used to



FIGURE 8 Textured CG model with environmental setting.

TABLE 1 Parameters of DJI Phantom 4 ProV2.0.

Max horizontal flight speed (position model)		**13.9 m/s**
Max wind speed		10 m/s
Camera resolution		5,472 × 3,648
Image sensor		1-in. CMOS, 20-M pixels
Camera lens	Focal length	8.8 mm (a 35-mm equivalent focal length of 24 mm)
	Aperture value	f/2.8–f/11

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construct another point cloud model in RealityCapture. As shown in Figure 9b,d, the values of the point cloud densities are different due to inevitable variations in lighting conditions and texture. Additionally, the absence of windows in the point cloud model of the as-built structure is attributed to reflection issues. Nevertheless, consistent tendencies are observed between the point cloud models of the as-built structure and the CG model. For example, some parts on the west side are missing, and the point cloud densities of the dented area on the surface of the building are lower than the prominent areas for both the as-built structure and the CG model. The similarity between the real and synthetic applications suggests that the assessment of the UAV flight path and camera settings can be carried out with the assistance of the GBDT. This approach can save significant time and effort that would have otherwise been spent in the field.

3.2 | Updated point cloud model of the structure

To explain the reason for the insufficient point cloud quality, the effect of various parameters such as different increment distances, camera angles, and distance between the UAV and building surfaces were examined in the synthetic environment. The point cloud models constructed on the synthetic images following different flight paths were evaluated for their visual integrity and point cloud densities. Results show that simply increasing the distance between the UAV and building surfaces could result in a significant improvement in the point cloud model. As shown in Figure 10, when the distance between the UAV and the building surface was above 10 m while the other parameters remained the same, a high-quality point cloud model was obtained. The reason for this improvement is that taking photos farther away from the building increases the

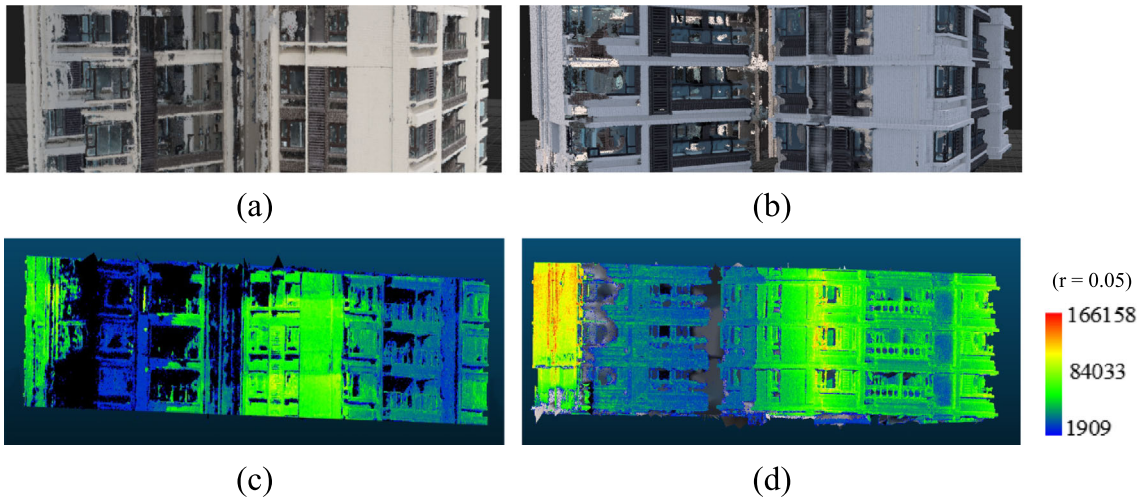


FIGURE 9 Initial point cloud models and point cloud densities of (a, c) as-built structure and (b, d) CG model.

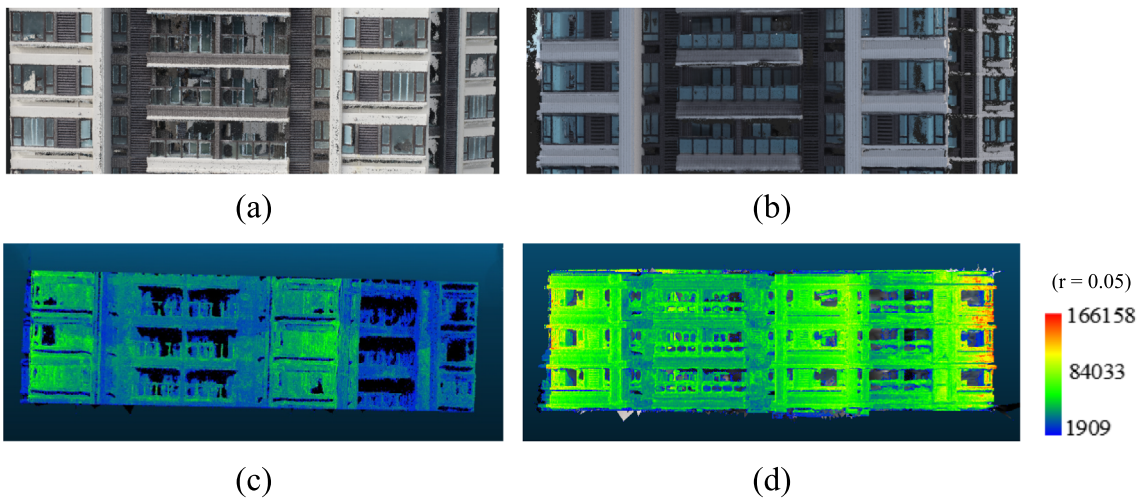


FIGURE 10 Updated point cloud models and point cloud densities of (a, c) as-built structure and (b, d) CG model.

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overlapping area between adjacent images, which benefits the coverage of the point cloud. Finally, this updated flight path was implemented on the as-built structure. The resulting point cloud model and point cloud density of the as-built structure were compared to those of the CG model, as shown in Figure 10.

3.3 | Dimension validation of the CG model

The high-quality point cloud model of the as-built structure, which was obtained in the previous section, is now being used as a reference to validate the dimensions of the CG model. The dimensions of the CG model are directly exported from the Blender (not from rendered images), and a quantitative assessment is made between the CG model and the as-built structure by Hausdorff distance. Hausdorff distance is a widely used performance measure in computer vision to calculate the distance between two point sets.[33] The two sets are close in the Hausdorff distance if every point of either set is close to some point of the other set. Therefore, the CG model is considered accurate with a large quantity of points located in the range of a small Hausdroff distance. Using the physical dimension as the scale for the Hausdroff distance, the proportion of points reaches 86.01% in the Hausdroff distance range less than 100 cm. This difference of 100 cm is equal to a percentage error of less than 4% in reference to the dimensions of the structure (25.0 m $\times$ 27.2 m $\times$ 138 m), which indicates a good agreement between the CG model and the as-built structure.

4 | DEVELOPMENT OF THE DAMAGED CG MODEL

The damaged CG model which serves as a testbed for UAV-aided structural inspection is built. To place more emphasis on potential damage locations, FE analysis is conducted to determine the damage locations under a specific earthquake. Those damage locations, referred to as damage hotspots, are then marked in the CG model with an additional texture layer.

4.1 | Seismic nonlinear analysis of the FE model

In practical applications, the actual excitation should be used to acquire damage information of the structure by FE analysis. For illustration purposes, the seismic ground motion of the 1940 El Centro earthquake is adopted as the excitation in this study. The peak ground acceleration (PGA) of the seismic excitation is deliberately scaled to a large value of 0.22 g so that significant damage can be detected in the structure. This PGA corresponds to the rare earthquake of intensity seven according to the Chinese seismic code.[34]

Time histories of structural responses are obtained through the nonlinear time history analysis under the seismic excitation. Interstory drift is an important index to evaluate the safety condition of structures. Cai et al.[35] suggested that interstory drift can be divided into harmless and harmful interstory drifts. The former is caused by the pure bending of the structure while the latter is caused by the shear reaction between the floors which is the main cause of damage in the structure. Figure 11a shows the maximum total and harmful interstory drift ratios (interstory drift divided by the height of the floor) at each floor of the structure under the scaled (0.22 g) El Centro earthquake. The interstory drifts may follow a different pattern under earthquakes with different frequency characteristics. The elastic limit of the interstory drift ratio (1/1000 according to the Chinese seismic code[34]) is marked with a vertical line as a reference. Note that this elastic limit is a conservative guideline for engineering design, and thus, damage is not expected on floors with interstory drift ratios below this limit. By referring to this limit, the maximum total interstory drift ratios (all exceed the elastic limit) suggest that all the floors have structural damage whereas the maximum harmful interstory ratios suggest that floors from 15 to 17 are not damaged. By checking the stresses of the shear walls, the concrete condition is found to be consistent with the harmful interstory drift ratios rather than the total interstory drift. Specifically, the shell stresses on floors 14–21 are less than the tensile strength of the concrete, indicating that the concrete is intact without any cracks. Therefore, the harmful interstory drift is used to determine the floors where damage is most likely to occur. Take floor 6 for example, Figure 11b shows the harmful interstory drift time history, 3.07 and 4.56 s (labeled as A and B, respectively) are marked with red dots for analysis in the subsequent section.

4.2 | Damage hotspot

To find the damage hotspots on those potentially damaged floors, the time histories of harmful interstory drifts and associated stress color maps are further examined. From Figure 11b, the two points in time with the largest magnitudes (i.e., 3.07 and 4.56 s) are selected to explore further. The stress color maps for the time point A (3.07 s) are shown in Figure 12, for example, where the color bar of the stress color map is adjusted so that the minimum tensile stress displayed is the same as the ultimate tensile strength (2.04 N/mm^2) of the concrete on those floors (floors 1–8).

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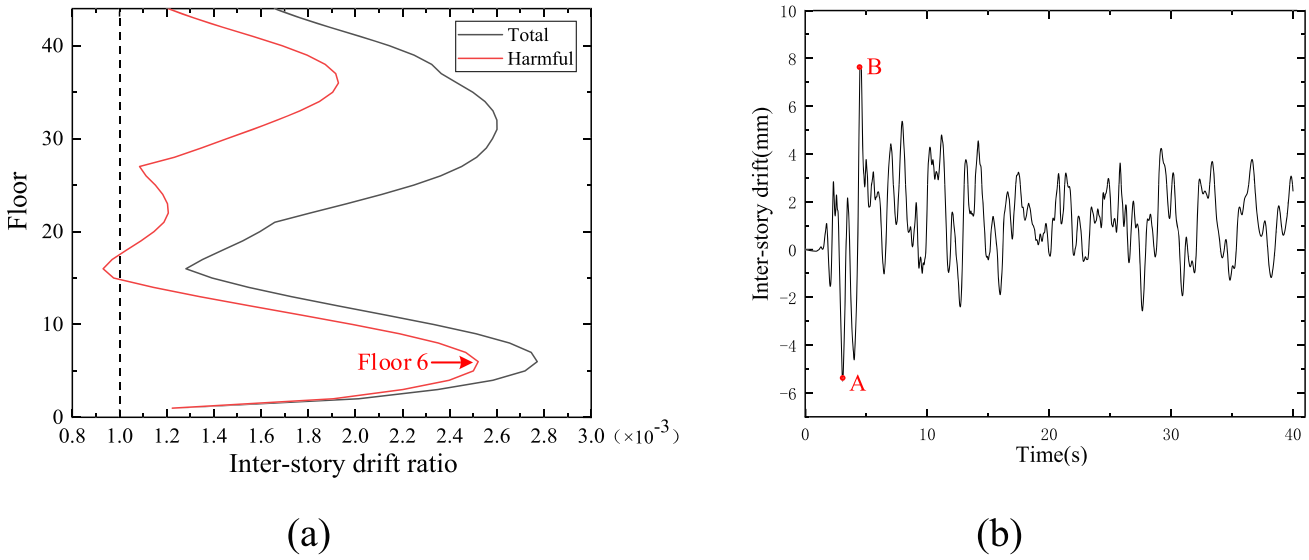


FIGURE 11 FE analysis results (a) maximum interstory drift ratios and (b) harmful interstory drift time history of floor 6. The vertical dashed line in (a) denotes the elastic limit of the interstory drift ratio (1/1000).

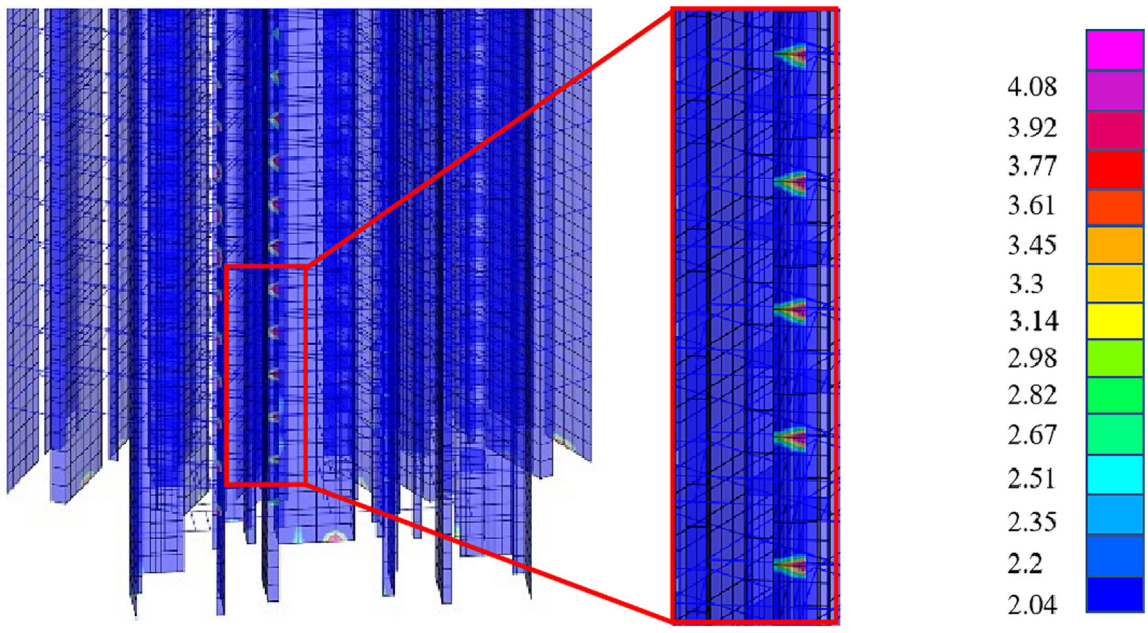


FIGURE 12 Stress color maps at Moment A. The zoom-in of the south external shear walls is marked with a red box.

Dark blue indicates that the stress is below the tensile limit and other colors are above the limit. The stress color maps show that the damage hotspots, suggested by the multi-color areas, occur in the south external shear walls of the structure. Those areas will be the focus of inspection in subsequent UAV flight planning.

4.3 | Damage texture of the CG model

In the CG model, the damage hotspots predicted by the FE model are textured with synthetic cracks, as shown in Figure 13. The damage texture is accomplished by superimposing a transparent layer containing the crack image (generated by design software based on the real seismic cracks on high-rise buildings) right above the original texture. The intention of the crack texture here is to mark the damage hotspots so that attention will be given to those areas in designing the UAV path. During extreme loading events, the locations of damage on both structural and

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nonstructural components tend to be generally consistent in terms of floor and orientation. Therefore, for time efficiency, the damage on nonstructural components and crack patterns describing the shapes and sizes of the cracks are not analyzed in this example. The realistic crack patterns and damage on nonstructural components can be obtained by component-level FE analysis using a finer mesh of the elements.

5 | GBDT-BASED POST-EARTHQUAKE UAV FLIGHT PLANNING FOR FIELD INSPECTION

To ensure the efficiency of the UAV-aided post-earthquake inspection in practical implementation, the synthetic environment of the GBDT is applied as the testbed to compare different inspection strategies. First, restrictions imposed by the physical environment including obstacles, weak satellite signal, wind speed, and lighting conditions are considered for the CG model. Based on these environmental considerations and the results of the FE analysis, the flight path and camera setting are designed. Subsequently, the designed flight schemes are carried out in the synthetic environment, and images rendered in the view of the UAV are obtained. Finally, different UAV flight schemes are evaluated, and the most appropriate scheme is determined.

5.1 | Environmental restrictions

For illustration purposes, non-ideal environmental conditions that may affect path planning and camera settings for the UAV are assumed, serving as an example of the necessary considerations in UAV flight planning.

Satellite signal interference can jeopardize UAV flight stability and accurate positioning, especially in complex urban environments. In this study, the target building is located a mere 15 m away from the twin building (143 m tall and situated on the west side of the target building). The satellite signal can be significantly disturbed in the area between the two adjacent high-rise buildings, which may increase the risk of flight-path failure and potential drone accidents. Moreover, strong winds resulting from the channeling effect between tall buildings can also destabilize UAV flights. The wind forces between parallel buildings can reach twice the normal level,[37] due to increased wind speeds in the narrow passage. Therefore, avoiding the narrow passage between the adjacent buildings when planning the flight path under strong winds is advisable. Additionally, the randomness of earthquakes requires UAV inspections to be carried out regardless of time, including times with poor lighting conditions such as dawn or dusk. In this study, poor lighting conditions akin to dusk are simulated in Blender using a sunlight strength of 500 W/m^2. The lighting condition will impact the selection of camera parameters.

While the aforementioned environmental restrictions are set for this specific example, adjustments can be easily made to explore the effects of various factors on path planning within the synthetic environment. For instance, simulating different lighting conditions can be achieved by modifying the light source and its intensity, introducing a dusty environment can be simulated by adding noise to the texture, and a significant change in wind speed can be replicated by randomly altering the pose of the camera in the synthetic environment.

5.2 | UAV flight schemes

Three UAV flight schemes are considered in this example. For different flight schemes, different combinations of the flight path and camera setting are considered.



FIGURE 13 Seismic damage: (a) crack photo[36] and (b) cracks on CG model.

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5.2.1 | Flight path

Two distinct flight paths, including a circular path and a parallel path, are considered for the high-rise structure (Figure 14). The two trajectories are inherently simple, as there is no need to account for collisions between multiple UAVs, and the presence of obstacles around the building is limited. The circular flight path takes the structure as the center of the circle. Such types of trajectory plans, as seen in Cheng et al,[38] provide a unified algorithm applicable to buildings of various sizes, ensuring sufficient coverage. In contrast, the parallel path is more intricate, as it adopts the structural surface with damage hotspots as a point of interest. Therefore, the normal distance from the UAV to those critical shear walls is set uniformly. The flight paths of different surfaces are connected by curved paths at the corners. Combining factors including obstacles, satellite signal, and wind speed, both flight paths are set to avoid the area between the target building and the adjacent building.

5.2.2 | Camera setting

Camera settings determine the quality of the images and 3D reconstruction of the structure. A synthetic movable camera consistent with the DJI Phantom 4 Pro V2.0 UAVused in the original survey is chosen for simulated image collection. In the Blender, the built-in camera uses the aperture parameters to adjust the camera exposure. Two very different aperture sizes, f/2.8 and f/5.6, are selected. The purpose is to illustrate that the quality of UAV-collected images can be evaluated through trials in the synthetic environment. For the physical camera, the exposure can be adjusted by changing the shutter speed, and the conversion between the aperture size of the synthetic camera and the shutter speed of the physical camera should be acquired for practical implementation. In the practical application, camera parameters need to be determined considering the UAV's distance from building surfaces and resolution requirements.

The working distance at 9 m from the structural surface is calculated from the field of view, camera resolution, and expected crack scale of 1 mm/pixel (Equations 1 and 2). The actual sensor size is 13.2 mm × 8.8 mm with the camera resolution considered. To ensure the 80% frontal and side overlap of collected images, the resulting intervals between two adjacent camera positions are 2.7 and 2.4 m in the horizontal and vertical directions, respectively.

$$HFOV = HRes \times ObjRes, \tag{1}$$

$$WD = \frac{HFOV}{SH} \times f, \tag{2}$$

where *HFOV* is the horizontal distance that can be captured by the field of view, *HRes* is the horizontal resolution of the camera, *ObjRes* is the object recognition resolution, *WD* is the working distance of the UAV relative to the structure, *SH* is the width of the camera sensor, and *f* is the focal length of the camera.

5.2.3 | Implementation in the synthetic environment

With different combinations of flight paths and camera parameters, the following three UAV flight schemes are compared in this study: (1) circular path with an aperture size of f/2.8, (2) parallel path with an aperture size of f/2.8, and (3) parallel path with an aperture size of f/5.6.

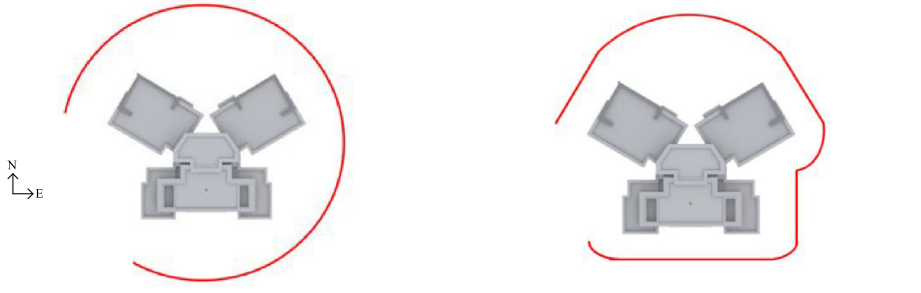


FIGURE 14 Two types of flight paths.

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In the synthetic environment, the flight path is created using the Bezier curve by editing pre-designed control points for the camera. Meanwhile, the virtual camera is constrained to track the center and surface of the high-rise structure for circular and parallel flight paths, respectively.

5.3 | Evaluation of UAV flight schemes

In this section, the three UAV flight schemes are evaluated to determine the most appropriate scheme for practical application. The resulting inspection strategy should meet the following requirements: (1) Photos of potentially damaged parts (damage hotspots) must be clear to determine the extent of damage on critical components. (2) The generated point cloud model of the overall structure must be accurate to judge the location and distribution of damage. (3) The UAV captures photos in as short a time as possible. It is worth mentioning that the time required to compare different flight paths using the GBDT primarily depends on the rendering of images from the CG model and the construction of the point cloud model. For the 45-floor high-rise building, the total time required for these two tasks is approximately one and a half hours. However, to assess the quality of photos and point cloud models, focusing on representative sections of the building (e.g., three to five floors) is sufficient. Consequently, the assessment time for one flight path can be limited to less than 10 minutes. Assessing different flight paths and choosing the appropriate one for the field inspection can therefore be implemented in a relatively short time.

5.3.1 | Photo of damage hotspot

The image sets are rendered using the Eevee engine[39] in the CG environment. Figure 15 shows examples of the images including the original views (top row) and zoom-ins on cracks (bottom row) collected following the three UAV flight schemes. Without distortion due to the skew angle, the shapes and sizes of the cracks in the images collected from Schemes (b) and (c) (parallel to the damaged building surface) can be better recognized than those collected from Scheme (a). However, due to the insufficient camera settings (less light reaches the sensor for the aperture size of f/5.6 than for the aperture size of f/2.8), the image of Scheme (c) is somewhat blurry.

5.3.2 | Accuracy of the point cloud model

The collected image set is also used for 3D point cloud model reconstruction in RealityCapture. The quality of the reconstruction is evaluated by comparing the point cloud models obtained by different flight schemes and the dimensions of the CG model directly exported from Blender. In contrast to the comparison in Section 3.3, where the ground truth is the as-built structure, the ground truth here is the CG model. It is important

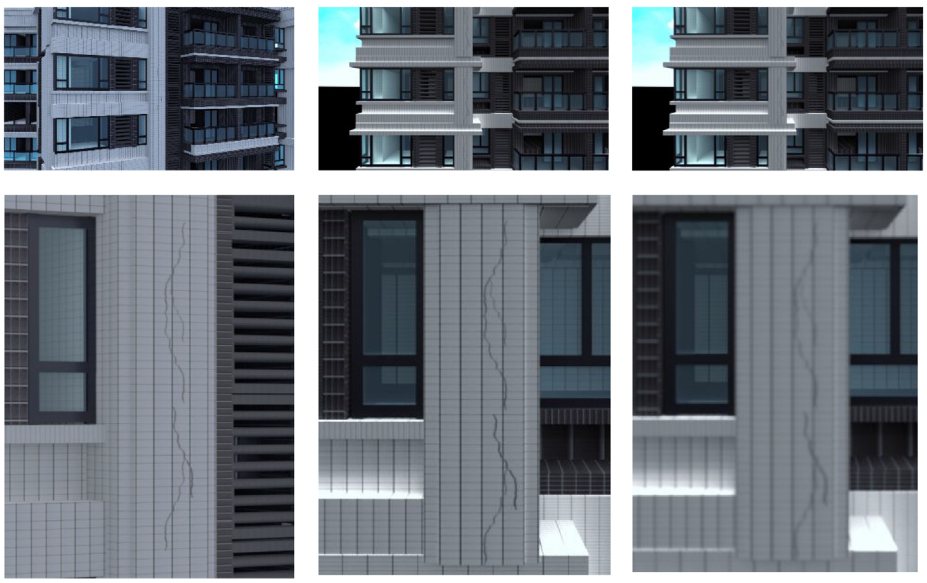


FIGURE 15 Images collected in the synthetic environment following different UAV flight schemes. Scheme (b) outperforms the others.

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to note that this CG model has been validated against the high-quality point cloud model of the as-built structure, as detailed in Section 3.3. The reconstructed model following each flight scheme is first registered with the CG model, and then, its point cloud distribution is used as the evaluation index. The reconstruction from Flight Scheme (c) is of low quality due to the suboptimal camera setting of the aperture size and thus not able to be registered with the CG model (Figure 16c). Table 2 shows the point cloud distribution in the Hausdorff distance range of less than 100 cm for the registered reconstructions of Schemes (a) and (b). In each 20-cm range, the reconstruction from Scheme (b) has a higher percentage of point cloud distribution than that of Scheme (a), indicating a more accurate reconstruction. The reconstruction quality is also visually compared in Figure 16a,b.

5.3.3 | Time of flight

The inspection time for potentially damaged structures in post-earthquake scenarios should be as short as possible as the battery life of most commercial UAVs is approximately 30 min.[32] The single-lap flight distance is approximated to be 146 and 124 m for the circular and parallel paths, respectively, and the number of laps covering all the damaged floors (floors 1–13 and 22–45) throughout the height of the building is 43. The speed of the UAV is assumed to be 3 m/s, a value lower than the maximum UAV speed of 5 m/s as adopted in Nagasawa et al.[40] This intentional reduction is implemented to ensure the safety of the flight, particularly under the challenging conditions of strong wind and poor lighting that are considered in the current example. Thus, the total time consumptions for the circular path (Scheme (a)) and parallel path (Schemes (b) and (c)) are 2093 and 1777 s, respectively. The flight schemes with the parallel path require a shorter time than that with the circular path, and the image collection using the parallel path can be completed within the battery life of the UAV. In the case where the inspection time is beyond the battery life of the UAV due to the large volume of the structure, change of battery should also be accounted for in the inspection time.

Based on the evaluation of the visual check, quality of the point cloud reconstruction, and time of the flight, Scheme (b) outperforms the others. This flight scheme is recommended for the assumed field UAV-aided post-earthquake inspection of the high-rise structure. It is important to note that the conclusion drawn is specific to this particular example. Alterations in environmental restrictions and UAV speed may result in a different optimal inspection scheme.

6 | FUTURE DEVELOPMENT

In the above section, the GBDT is devised to test and evaluate post-earthquake UAV flight schemes based on the knowledge of damage location and environmental restrictions. While the assumptions and conditions employed have been simplified, this example nonetheless demonstrates the capability of GBDT for use in the field. Applications of the GBDT toward automated detection of damage and assessment of structural conditions will be the focus of research in the future.

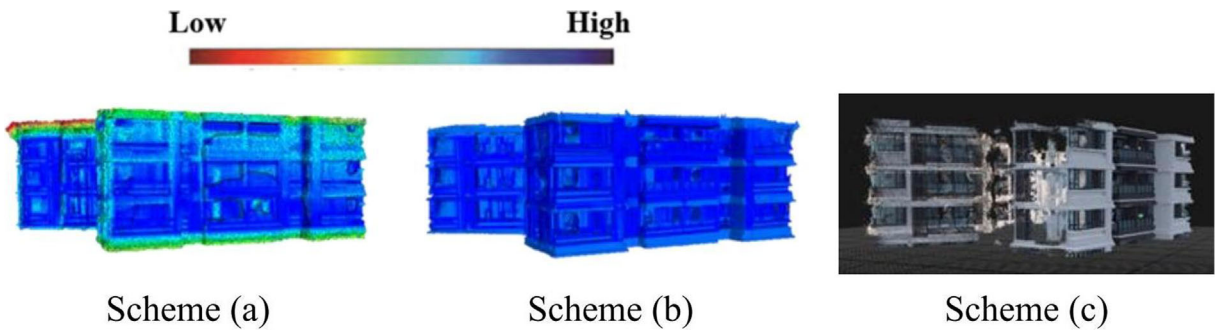


FIGURE 16 Point-cloud reconstruction for the three UAV flight schemes.

TABLE 2 Point cloud distribution of different Hausdorff distance ranges.

	Hausdorff distance range (cm)				
Reconstructed model	**0–20**	**20–40**	**40–60**	**60–80**	**80–100**
Scheme (a)	66.86%	5.00%	3.96%	3.29%	2.89%
Scheme (b)	70.24%	8.24%	7.35%	5.57%	4.95%

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6.1 | Component-level FE analysis

In this study, only the global FE analysis of the high-rise structure is conducted and the CG model with damage hotspots is constructed accordingly. Extending the previous work about PBGM,[18,28] the responses of the high-rise structure from the global FE analysis will be used as the boundary conditions to carry out a more detailed study on the structural and nonstructural components. The damage patterns resulting from the finer component-level FE analysis will be used to render realistic damage textures including cracks, spalling, and exposed rebars for the CG model.

6.2 | CG model construction

Because the nonstructural components and the building materials of high-rise structures for a certain function are limited to just a few types (e.g., apartment and office), a library of building parts and textures should be established. In addition, CG algorithms will be developed to obtain ready-to-use damage textures from the stress color maps of both global and component-level FE analyses. Consequently, the whole building process of the CG model will be automated to further raise the modeling efficiency, especially when creating the GBDT for a community of buildings.

6.3 | UAV path planning

In addition to guiding UAV path planning, the damaged hotspots identified by the FE model of the GBDT can be utilized to further optimize the inspection strategy. A detailed examination of the damaged areas, achieved by capturing additional photos, can enhance the accuracy of damage detection and evaluation. Furthermore, while UAV paths were tested for inspecting a single building in this study, the potential for multiple UAV path planning for a cluster of nearby buildings, as demonstrated in Nagasawa et al,[40] can be implemented. The quality of UAV-collected images can then be evaluated with the assistance of the GBDT.

6.4 | Damage detection

Given the ease with which UAV inspections can be conducted in the synthetic environment, automated scene mapping and feature recognition using advanced machine learning, as discussed in Kerle et al,[41] can also be conveniently implemented based on the virtually realistic images collected by the fictional UAV. The GBDT will be applied to test and train machine learning algorithms for automated detection of damage and assessment of structural conditions. In particular, the CG model with detailed damage patterns (a realistic visual representation of the as-built structure) will be used to generate synthetic data for the purpose of seismic damage detection and identification.[42,43] The information obtained from the visual damage will then be trained to infer the structural condition after the earthquake, making the post-earthquake inspection and evaluation of structures more intelligent and efficient.

7 | CONCLUSION

In this study, a GBDT framework is developed for UAV-aided post-earthquake inspection of high-rise buildings and validated using a real-world high-rise building in Guangzhou, China. The GBDT is comprised of an FE model and a CG model, with the former simulating the physical properties and the latter being the photorealistic virtual representation of the as-built structure. The virtual representation of the structure combined with environmental considerations from the physical world provides a synthetic testbed in which every step of the post-earthquake inspection procedure can be planned and evaluated virtually.

To avoid the cumbersome nature of constructing the numerous components in the CG model of the high-rise building, structural components are automatically imported from the FE model, whereas the nonstructural components are added according to the architectural drawings. Visual checks and quantitative comparisons using the point cloud distribution are made between the CG model and the as-built structure. The high consistency found between the CG model and its physical counterpart validates the fast modeling process as well as the accuracy of the virtual representation.

To demonstrate the efficacy of the GBDT framework in assisting practical applications, the validated GBDT of the high-rise building is used to evaluate UAV flight schemes for post-earthquake inspection. To place more emphasis on potential damage, FE analysis is conducted to determine the damage locations under the earthquake, and consistent damage hotspots are then marked on the CG model. Environmental restrictions such as obstacles, weak satellite signals, wind speed, and lighting conditions are also considered, further enhancing the authenticity of the

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synthetic environment. Within this environment, three UAV flight schemes with different flight paths and camera settings are implemented virtually. The evaluation of photo quality, accuracy of point cloud distribution, and time consumption allows for a comprehensive comparison of the different flight schemes. Based on these evaluations, the best scheme for the assumed field inspection is determined to be the parallel path with the greater camera exposure.

These results highlight the significant potential of the GBDT framework as a valuable tool in guiding real-world post-earthquake inspection activities. Its ability to simulate and evaluate various scenarios contributes to efficient decisionmaking and improves the overall effectiveness of structural inspections in the aftermath of earthquakes.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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